Running the Theory: An Empirical Study of Model-Based Reinforcement Learning for Confounded POMDPs

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Abstract

Offline reinforcement learning is systematically biased when the behavior policy acted on latent state the learner never observes. Hong et al. (2024) give a model-based theory for this setting, identifying reward and transition models through two families of *bridge functions* and planning pessimistically over a confidence region — but the paper is theory only: no algorithm, no code, no experiments. We build the pipeline end-to-end, verify it against exact ground truth to machine precision, and report what running it reveals. Run without the bridge-class bound we omitted, our pessimism step diverges to -21 against a true value of 2.02; restoring the class caps the excursion at -3 without removing it, and a confounding-blind plug-in is never beaten at any width — the pessimism layer costs selection accuracy and returns nothing. At benchmark scale a confounding-blind baseline beats every bridge-based method in all 15 sweep cells, worst at *zero* confounding — estimation variance, not confounding bias, is the binding constraint. Pessimistic *selection* fails for a reason we establish by sweeping it: the penalty tracks behavior-policy coverage, and driving coverage of the optimal action up drives regret to exactly zero. A negative-control quality sweep then quantifies how good the framework's central assumption has to be.

1. Introduction

Offline RL from logged data is unreliable when the logging policy observed more than the learner does. In a partially observable MDP whose *latent* state drives both the recorded actions and the outcomes, the data are *confounded*: treating the observation as the state yields a bias that does not vanish with more data. This is the central obstacle in healthcare and operations, where unrecorded drivers of decisions are the norm.

Hong et al. (2024) address exactly this setting. Using a pre-decision *negative control* O_0, they identify the confounded POMDP's reward and transition models through two families of bridge functions (b_R, b_D), then plan pessimistically over a policy-independent confidence region. The result is elegant and, importantly, stated for general spaces: states and observations may be continuous, and general function approximation is not merely permitted but required. The paper is, however, *theory only* — it contains no algorithm, no code, and no experiments. Its neighbors are likewise unrealized empirically for this exact pipeline: the closest follow-up runs the bridge *estimator* in isolation (Venkatesh & Malikopoulos, 2026), and the pessimism precedent is theory-only and for the *un*confounded case (Guo et al., 2022).

This leaves a concrete question that only implementation can answer: *when the framework is built end-to-end, does model-based bridge identification actually de-bias policy evaluation at practical sample sizes, and can its abstract pessimistic* max–min *step be made to run at all?*

Contributions. [1]

1. An end-to-end implementation estimating *both* bridge families and carrying them through to a tractable pessimistic policy-*selection* step, verified against exact oracles to $\sim 10^{-15}$ (Section 4.1) — a capability claim we can check ourselves, not a primacy claim.
2. A *diagnosis* of a failure mode in coupled multi-block pessimism run without the class bound, plus a signal-subspace heuristic that restores in-range values but, by our own test, never beats the plug-in selector — a negative result (Section 3.3, Section 4.3).
3. A characterization of *where the approach stops work-*

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[1]Code and raw results: `github.com/soheilsayahvarg/model-based-rl-confounded-pomdps` (private repository; access on request). The benchmark builds on the public `gumbel-max-scm` simulator (Oberst & Sontag, 2019).

ing: a naive baseline wins all 15 cells of a benchmark sweep, worst where there is nothing to correct (Section 4.4).

4. A kernel extension to continuous spaces, and — in the paper-faithful configuration with finite actions — a diagnostic identifying *behavior-policy coverage* as the mechanism behind pessimistic selection failure (Section 4.5, Section 4.6).

2. Related Work

Proximal causal inference. Bridge functions originate in the double negative control literature (Miao et al., 2018; Tchetgen Tchetgen et al., 2020), where a pair of proxy variables identifies a causal effect under unmeasured confounding. Mastouri et al. (2021) give the kernel two-stage estimation procedure our continuous extension instantiates.

Model-free OPE in confounded POMDPs. Shi et al. (2022) and Bennett & Kallus (2024) identify policy values through *value* bridge functions solved from Bellman-type backward moment equations. These depend on the evaluation policy, so a new estimation problem is solved per policy; the model-based route of Hong et al. (2024) instead extracts policy-independent reward and transition information, which is what makes a policy-independent confidence region possible. We implement Shi et al. (2022) as a competing baseline.

Pessimism in offline RL. Conservative planning over confidence regions is standard in fully observed offline RL and has a POMDP precedent in Guo et al. (2022), which is theory-only and unconfounded. The failure mode of Section 3.3 is specific to the interaction between a structural null space and *joint* multi-block minimization.

3. Method

3.1. Identification

We adopt the setting of Hong et al. (2024): a finite-horizon POMDP with latent states S_t, observations O_t, actions A_t and rewards R_t, where the behavior policy may depend on S_t.

Assumption 3.1 (Valid pre-decision negative control)**.** $O_0 \perp\!\!\!\perp (O_t, O_{t+1}, R_t) \mid S_t, A_t, H_{t-1}$.

Reward-emission and dynamic-emission bridge functions solve the conditional moment restrictions

$$\mathbb{E}\big[b_R^{[t]}(A_t, O_t, r_t, o_t) \mid H_{t-1}, A_t, O_0\big] = p(r_t, o_t \mid \cdot), \quad (1)$$

and analogously for $b_D^{[t]}$ with target $p(o_{t+1}, o_t \mid \cdot)$. Under a completeness condition, the policy value of *any* history-dependent policy is identified by a sequential contraction of these bridges (Hong et al., 2024, Thm. 3.5). We implement this contraction exactly and verify it against dynamic programming on the latent model.

3.2. Estimation

Stage 1 fits a conditional mean embedding $\hat{\mu}_{W_t|x}$ by kernel ridge regression; Stage 2 solves the resulting empirical operator equation in closed form, $\alpha = (G + N_2\lambda_2 I)^{-1}\hat{p}$. We implement both the dual (representer) and primal (operator) forms and verify they agree to 2.4×10^{-15}, which catches derivation errors that no single implementation could.

Regularization does not transfer. The paper's RKHS rate schedules, applied literally to a tabular instantiation, over-shrink the estimator by $\approx 80\%$ by the third time-step. We grid-calibrate instead: a 0.7:0.3 split with $\lambda_1 = 1/N_1$ and $\lambda_2 = 0.03/\sqrt{N_2}$. These values are measured, not derived.

The usual ill-posedness diagnostic misleads. Because the observation space is structurally richer than the latent space ($|O| > |S|$), the design matrix carries a null space that pins $\mathrm{cond}(G)$ flat even as estimation quality degrades tenfold — confirmed by an O_0-noise sweep in which Stage-1 error grows $10\times$ while $\mathrm{cond}(G)$ does not move. We monitor the smallest *signal* eigenvalue instead.

3.3. Pessimism, its failure mode, and a repair

Confidence regions are exact ellipsoids centered at $\hat{b}$, and the per-block inner minimum is the standard ellipsoid support-function identity, $V(\hat{b}) - \sqrt{\xi}\,\|g_\pi\|_{H^{-1}}$. This is not new mathematics and we do not claim it as a contribution. The difficulty is the paper's *coupled* minimization across blocks.

Failure mode. Run class-free, the joint minimization exploits precisely the null-space directions identified above: they cost the confidence-region geometry almost nothing, so the optimizer walks arbitrarily far into them and the pessimistic value diverges (Section 4.3). The diagnosis and the spectral-monitor fix share one root cause.

Signal-Projected Pessimism. We restrict the pessimistic search to the empirically identified signal subspace via an SVD/eigengap decomposition with orthonormal basis U and $H_{\mathrm{sub}} = U^\top H U$. This restores informative values and correct selection — but, as Section 4.3 shows, at a coverage cost we quantify rather than assume away.

3.4. Continuous extension

The framework is not inherently tabular: Hong et al. (2024) assume continuous S and O and call function approximation *inevitable* there. We instantiate the same pipeline with a two-stage **kernel** estimator on a linear-Gaussian confounded POMDP admitting an exact closed-form oracle, reusing the pessimism engine unmodified since it works purely in coefficient space (derivation: Appendix B).

4. Experiments

Multi-seed results use 5 seeds with 95% CIs, except Appendix I (3 seeds). Environments: a minimal tabular POMDP ($|S|=2, |A|=2, |O|=3, T=3$) with closed-form oracle bridges; a 720-observation benchmark built on the public `gumbel-max-scm` clinical simulator (Oberst & Sontag, 2019) with a manufactured O_0 satisfying the identification assumption by construction; and the continuous linear-Gaussian POMDP of Section 3.4. Everything runs on a laptop in NumPy; no GPU.

4.1. Correctness

We first pin the implementation against exact ground truth. The policy-value chain matches exact dynamic programming to 4.4×10^{-16}; the estimator matches the oracle bridges in the population limit to 3.0×10^{-15}; the dual and primal solves, independent derivations of the same estimator, agree to 2.4×10^{-15}; analytic gradients match finite differences to 4.9×10^{-10}. The dual-versus-primal check is the most informative, since it rules out derivation errors a single implementation would silently carry. This is what licenses reporting the negative results below as findings rather than bugs.

4.2. De-biasing works where identification is clean

On the toy environment the plug-in evaluator beats the confounding-blind baseline in **12/12** (N, π) cells. Worst-policy plug-in bias shrinks $0.104 \to 0.053 \to 0.036$ as N grows from 5k to 50k, while naive bias stays flat at 0.13–0.33 regardless of N: the textbook signature of removed versus irreducible bias.

4.3. The pessimism step, run class-free, diverges

Our first implementation minimised over the confidence region alone, omitting the bridge-class intersection Hong et al. (2024) carry through their Theorem 4.2. Run that way, the support-function minimum of Section 3.3 drives the best-policy V_{low} from 1.272 at c=0.03 through 0.123 and -3.235 to $\mathbf{-20.98}$ at c=1.0, against $V_{\text{true}} = 2.018$; regret rises $0.000 \to 0.513$. Restoring the class bounds the excursion without removing it — Hong et al.'s Assumption D.16(d) ℓ_2 ball holds V_{low} to -3.026, a 6.9× shrinkage, selections unchanged. Signal-Projected Pessimism reaches 1.338 with 0.000 regret at c=0.03, but so does *every* variant there: our earlier advantage was an artifact of reading across region sizes, and its real gain is confined to mid-grid. A confounding-blind plug-in achieves 0.000 regret at every width and is **never beaten** (Appendix I).

Table 1. Benchmark sweep (5 confounding strengths × 3 sample sizes × 5 seeds). MAE against exact ground truth.

Method	MAE range	Beats naive
Naive (confounding-blind)	**0.013–0.074**	—
Model-based (plain)	0.072–0.230	0/15
Model-based (shrunk)	0.063–0.240	0/15
Model-free proximal	0.164–0.180	0/15

The honest limit. Stress-testing our own fix, the true bridges leak up to **21% of their norm** outside the projected subspace, so that region provably does *not* contain the truth: its honest coverage is zero. The trade-off is **coverage against informativeness** — vanilla covers the truth but says nothing; the projection is informative and empirically correct but uncertified, with $V_{\text{low}} \le V_{\text{true}}$ holding by sign-alignment in every test rather than by guarantee.

4.4. At scale, a naive baseline wins

The de-biasing method's *worst* performance is at $\kappa = 0$, where it is 2.8–6.8× worse than naive — precisely where there is nothing to correct. That is the clearest possible signature that the dominant error source at this scale is **estimation variance**, not the confounding bias the method targets: inverting many small, noisy per-cell matrices injects more noise than the bias it removes. The model-free comparator is flat in N, which we confirmed is an identification limit of the benchmark's binary proxy rather than a bug — it *is* consistent on the toy (0.0076 max error at N=200k).

4.5. Continuous state, action and observation spaces

The kernel instantiation reproduces the tabular story. Oracle self-consistency is 1.665×10^{-16}, the gradient check $\approx 10^{-11}$, bridge recovery improves with sample size ($0.220 \to 0.055$ reward, $0.240 \to 0.102$ dynamics, for $N : 300 \to$ 4,000), and plug-in de-biasing beats naive in **4/4** policies by 3–11× in absolute bias — the continuous analogue of the tabular 12/12, with the same signature: naive bias tight but systematically wrong, plug-in bias smaller but noisier (per-policy table in Appendix F).

Pessimism validity holds throughout, but **validity is not selection quality**: ranking by V_{low} is consistently worse than ranking by the plug-in value (regret 0.40–0.48 versus 0.241), unlike the tabular case's clean 0.000. Two bottlenecks are regime-specific: dense kernel ridge costs $O(N_1^3)$, capping $N \approx$ 4,000; and a *finite-dimensional* kernel silently re-creates the rank deficiency of Section 3.3, caught only by checking the Gram matrix's rank (Appendix B).

Table 2. Coverage sweep. Varying the logging policy's dose preference changes *only the data* — every candidate's true value is invariant and asserted so at each grid point. Regret is averaged over $c \in \{0.1, 0.3, 1.0\}$, 5 seeds. "Ratio" is the optimal policy's pessimism penalty divided by the mean penalty of the others.

Coverage of optimal policy	Pessimistic regret	Plug-in regret	Penalty ratio
0.124	0.367 ± 0.051	0.000	1.85
0.244	0.211	0.000	1.40
0.407	0.211	0.000	1.04
0.582	0.140	0.000	0.73
$\mathbf{0.727}$	$\mathbf{0.000}$	0.000	$\mathbf{0.49}$

4.6. The paper-faithful configuration, and why pessimistic selection fails

Hong et al. (2024) assume continuous S and O but an explicitly *finite* action space, so Section 4.5 varies two things at once. We repeated the study with $A = \{0, 0.5, 1\}$, giving each action level its own bridge coefficient (a delta kernel on the action coordinate tensored with a linear kernel on the observation), all else fixed. The oracle check is exact (0.000), recovery is monotone in N, and validity again holds at 100%.

The anomaly persists, refuting our own hypothesis that continuous actions caused it. It is also not an optimization failure: restart gaps here are $\leq 4 \times 10^{-16}$ up to c=1, so coordinate descent is finding the global optimum. A diagnostic instead found the pessimism penalty to be monotonically inverse to behavior-policy coverage, with the optimal policy the least covered (12% of logged actions, against 62% for the worst policy).

We turned that into a prediction and tested it. If coverage is the mechanism, raising the logging policy's coverage of the optimal action should drive pessimistic regret to zero. Table 2 shows exactly that: correlation between coverage and regret is -0.944, regret falls from 0.367 to **exactly zero**, and the penalty ratio crosses 1 near 43% coverage — below it the optimal policy is penalized *more* than the alternatives, above it *less*. Plug-in selection is perfect throughout, the control confirming that estimation quality is not what moves.

Pessimism is therefore not malfunctioning: it answers "which policy can I most confidently guarantee?", which coincides with "which is best?" only under adequate coverage. De-biasing likewise wins $2/4$ rather than $4/4$ — where naive bias is negligible the correction only adds variance.

4.7. How good does the negative control have to be?

Theory cannot say what happens when the negative-control assumption only approximately holds, since it is what the theory conditions on; a simulator with a quality knob can. We sweep two failure modes (Appendix D). Raising the control's noise σ_0 leaves Assumption 3.1 true and degrades only informativeness: the plug-in's systematic-bias advantage survives to $\sigma_0 \approx 0.5$ and is gone by 1.0. Leaking that noise into the reward ($r_t \mathrel{+}= \delta\, \epsilon_0$) instead makes the assumption *false* while leaving $\mathbb{E}[R]$ — and the oracle — intact; the advantage is gone by $\delta = 0.5$. Violating the assumption costs more than weakening the instrument, now quantified rather than assumed, though the two knobs are not on a common scale so the ordering, not the ratio, is what transfers.

One caution: recovery error is a poor proxy for value bias. At δ=1 it is $9\times$ baseline while systematic value bias grows $1.9\times$.

5. Limitations

The pessimism guarantee is empirical. The projection provably excludes part of the truth (21% norm leakage), so $V_{\text{low}} \leq V_{\text{true}}$ is an observation, not a theorem.

The coverage explanation is tested in one environment. Table 2 confirms the prediction, but varies coverage through a single channel in one linear-Gaussian POMDP; absent actions and state-dependent gaps are untested. With four candidate policies regret takes few discrete values — two grid points tie — and $r = -0.944$ rests on n=5 points, so we lean on the manipulation, not the coefficient.

Restricted scope. Candidate policies are affine and observation-only, or observation-independent — what preserves the exact oracle. All use T=3; the *continuous* ones scalar spaces and $N \leq 4{,}000$, the tabular and benchmark runs up to $N = 200{,}000$. Our one confounding-aware comparator is under-identified by the benchmark's binary proxy, so at that scale *no* identifying confounding-aware method is in the comparison — Section 4.4's "naive wins" is bounded by that.

Statistical self-correction. Two earlier headline numbers rested on 3-seed deviations; at 5 seeds one changed materially and the other was refuted. Both are in Appendix E, with the retracted mechanism for Section 4.6.

6. Conclusion

Running a theory-only framework end-to-end showed what the theory did not: our class-free pessimism diverges and the class bound caps but does not cure it, a naive baseline dominates at scale through variance rather than bias, and pessimistic selection tracks coverage — a manipulable cause, and the result we would build on next.

References

Bennett, A. and Kallus, N. Proximal reinforcement learning: Efficient off-policy evaluation in partially observed Markov decision processes. *Operations Research*, 2024.

Caponnetto, A. and De Vito, E. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, 2007.

Guo, H., Cai, Q., Yang, Z., and Wang, Z. Provably efficient offline reinforcement learning for partially observable Markov decision processes. In *International Conference on Machine Learning (ICML)*, 2022.

Hong, M., Qi, Z., and Xu, Y. Model-based reinforcement learning for confounded POMDPs. In *International Conference on Machine Learning (ICML)*, 2024.

Mastouri, A., Zhu, Y., Gultchin, L., Korba, A., Silva, R., Kusner, M., Gretton, A., and Muandet, K. Proximal causal learning with kernels: Two-stage estimation and moment restriction. In *International Conference on Machine Learning (ICML)*, 2021.

Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993, 2018.

Oberst, M. and Sontag, D. Counterfactual off-policy evaluation with Gumbel-max structural causal models. In *International Conference on Machine Learning (ICML)*, 2019.

Shi, C., Uehara, M., Huang, J., and Jiang, N. A minimax learning approach to off-policy evaluation in confounded partially observable Markov decision processes. In *International Conference on Machine Learning (ICML)*, 2022.

Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. An introduction to proximal causal learning. *arXiv preprint arXiv:2009.10982*, 2020.

Venkatesh, V. and Malikopoulos, A. Cross-fitted proximal learning for model-based reinforcement learning. *arXiv preprint arXiv:2604.05185*, 2026.

A. Experimental Setup

Environments. *Toy POMDP:* $|S|$=2, $|A|$=2, $|O|$=3, T=3, with closed-form oracle bridge functions obtained by pseudo-inverse and exact dynamic-programming policy values. This is the only substrate where recovery error against ground truth is directly measurable, and it carries every machine-precision correctness check.

Benchmark: 720 observable-core states $\times$ 2 latent states, 8 actions, T=3, built on the public `gumbel-max-scm` clinical simulator (Oberst & Sontag, 2019). The latent index drives dynamics and behavior but is censored from the logs. We construct a pre-decision negative control O_0 satisfying the identification assumption by construction, plus a per-step noisy marker that guarantees bridge existence — deterministic masking alone gives rank $720 < 1440$, i.e. no valid one-step bridge. A confounding knob $\kappa \in \{0, 0.25, 0.5, 0.75, 1.0\}$ sweeps from no confounding to strong confounding.

Continuous: scalar linear-Gaussian confounded POMDP, described in Appendix B.

Protocol. All multi-seed results use seeds $\{0, 1, 2, 3, 4\}$ and report $1.96\,\hat{\sigma}/\sqrt{n}$ intervals. Ground truth throughout is exact dynamic programming on the latent SCM (tabular) or a closed-form recursion (continuous). The full pipeline runs on a laptop in NumPy; no GPU is used or required.

Scaling the benchmark estimator. A direct implementation of the paper's history-conditioning would require on the order of billions of table cells at benchmark scale. Three exact structural factorizations specific to the environment's observation design — hashed encoding, core-diagonal bridge support, and closed-form 2×2 local inversions — keep the pipeline laptop-scale without approximation.

B. Continuous Extension: Derivation and Details

B.1. Environment

State, action, observation and reward are all real-valued. The observation is a noisy proxy $o_t = s_t + \varepsilon_t$ for the hidden state, and the logging policy may act on the true s_t (confounded) or only on o_t (not confounded), mixed by a knob κ. This is the classical *measurement-error* setting that proximal causal inference was designed for: fitting a model that treats the noisy proxy as if it were the state induces errors-in-variables bias whenever $\kappa > 0$.

Because the reward is linear and candidate policies are affine and observation-only ($a = Ko + c$), the closed-loop mean trajectory is an exact scalar recursion — no covariance propagation is needed. This yields an exact closed-form oracle for both policy values and bridge coefficients, verified to reproduce each other to 1.665×10^{-16}.

B.2. Mean-value bridges

The paper's bridge functions are *distributional*: they must match $p(y \mid X_t)$ for every y. Reproducing that in continuous y requires conditional density estimation. Since the downstream quantity is a policy-value *mean* and the pessimism machinery concerns estimation uncertainty in the bridge coefficients rather than the environment's intrinsic reward noise, a conditional-mean moment restriction suffices and is exact for that purpose:

$$\mathbb{E}[b(A_t, O_t) \mid X_t = x] = \mathbb{E}[Y_t \mid X_t = x] \quad \forall x, \tag{2}$$

with $Y_t = R_t$ (reward bridge) or O_{t+1} (dynamics bridge). This is exactly a nonparametric instrumental-variable regression, with $W = (A, O)$ confounded, $X = (A, H_{t-1}, O_0)$ the instrument, and Y the outcome — well precedented in the proximal literature (Mastouri et al., 2021). It is a deliberate scope reduction, not a fundamental barrier.

B.3. Two-stage kernel solve

We derived the closed form directly from a stated objective rather than pattern-matching a remembered formula, because a mis-remembered index convention would silently produce a wrong-but-plausible estimator — the exact failure mode oracle verification exists to catch. With K_X, K_W the instrument- and bridge-side kernels,

$$\Gamma = (K_X(X_1, X_1) + N_1 \lambda_1 I)^{-1} K_X(X_1, X_2), \tag{3}$$

$$Z = K_W(W_1, W_1)\,\Gamma, \tag{4}$$

$$H = \tfrac{1}{N_2} Z Z^\top + \lambda_2 K_W(W_1, W_1), \quad g = \tfrac{1}{N_2} Z y_2, \tag{5}$$

and $\beta = H^{-1}g$. **Unifying check:** when K_W is a one-hot/delta kernel (the tabular case), $K_W(W_1, W_1) = I$ exactly and this reduces to the tabular estimator's own primal solve — a correctness check on the derivation itself, not merely an analogy.

B.4. A rank-deficiency pitfall

For the primary verified path we take K_W to be a *linear* kernel, since the true bridges are exactly linear and a linear kernel's RKHS therefore contains them. But a linear kernel has a finite (2-dimensional) feature space, so the $N_1 \times N_1$ Gram matrix $K_W(W_1, W_1)$ has rank ≤ 2 — confirmed empirically, with the remaining eigenvalues at the $\sim 10^{-13}$ floor. The dual system is therefore numerically meaningless in those directions.

Notably, the specific projected quantity we originally extracted ($\hat{\theta} = W_1^\top \beta$) turned out to be *invariant* to that instability and produced identical recovery numbers before and after the fix — so the bug was invisible from the recovery metrics. It still had to be fixed, because H and $\hat{b}$ enter the pessimism ellipsoid directly, where no such invariance protects them. The fix is to solve in the 2-dimensional *primal* coordinates: mathematically identical, exactly well-conditioned, and incidentally $\sim 1000\times$ cheaper.

B.5. Spectral monitoring

The tabular fix for a misleading condition number was a hard eigengap on the smallest signal eigenvalue, correct there because the null space is *structural* ($|O| > |S|$, exactly zero signal beyond it). A continuous kernel Gram matrix has smooth spectral decay rather than a structural wall, so a hard eigengap does not transfer. We use the standard kernel-ridge *effective dimension* $N_{\text{eff}}(\lambda) = \operatorname{tr} K(K + N\lambda I)^{-1}$ (Caponnetto & De Vito, 2007) instead.

B.6. Gradients

The value chain is multilinear in each stage's coefficients, so reward-block gradients are exact. A dynamics block's coefficients additionally shift the *evaluation point* of every later stage, and that dependence runs through the kernel nonlinearly; we keep the coefficient-linear part exact and obtain the one remaining scalar sensitivity by a single 1-D central difference — $O(T)$ extra work per gradient call, not $O(N_1)$ — verified against numerical differentiation of the whole chain to $\approx 10^{-11}$.

B.7. Regularization: a tested-and-rejected hypothesis

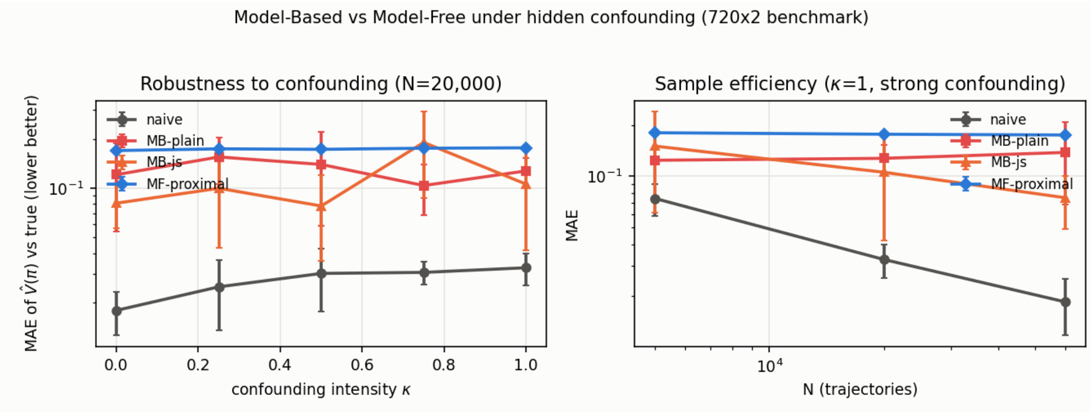
We hypothesized that the tabular-calibrated $\lambda_1 = 1/N_1$ would be too weak for a genuine kernel conditional mean embedding, since the measured effective dimension came out close to N_1 itself. We tested $1/\sqrt{N_1}$ and $5/\sqrt{N_1}$ with 5-seed averaging: both performed *worse*. We kept $1/N_1$. This is the mirror image of the tabular finding — there, the paper's RKHS schedule failed to transfer to a tabular setting; here, the tabular-calibrated schedule transferred fine to a genuine kernel.

B.8. Full pessimism sweep

c	0.03	0.1	0.3	1.0	3.0
Fraction valid ($V_{\text{low}} \leq V_{\text{true}}$)	1.00	1.00	1.00	1.00	1.00
Pessimistic regret	0.479	0.438	0.397	0.397	0.397
Plug-in regret	0.241	0.241	0.241	0.241	0.241
Max restart gap	2.2e-16	1.36	5.26	20.4	62.2

Validity holds everywhere. Selection quality does not match the tabular case, and the restart gap grows sharply with region width — consistent with the optimization landscape becoming genuinely harder to solve exactly, rather than merely more conservative.

C. Benchmark Sweep Figure



Naive vs. model-based (plain/shrunk) vs. model-free MAE, 5-seed 95% CIs. Left: vs. confounding strength κ at N=20,000. Right: vs. sample size at κ=1. Naive stays lowest across both panels. The wide, overlapping error bars on the two model-based variants are the visual evidence behind the retracted shrinkage-variance claim of Appendix E.

D. Negative-Control Quality: Full Sweeps

Two metrics are reported at every grid point, because they disagree by construction and reporting only one inverts the conclusion: *systematic bias* is the mean over policies of | mean over seeds of $(\hat{V} - V)$|, isolating what the method is designed to remove; *total error* is the mean over policies and seeds of $|\hat{V} - V|$, which is what a practitioner pays and includes the variance the correction costs. The plug-in wins the first and loses the second. N=2,000, 5 seeds.

[N1] Strength: σ_0 varies, Assumption 3.1 holds throughout

		Systematic bias		Total error	
σ_0	Recov. err	plug-in	naive	plug-in	naive
0.10	0.074	**0.027**	0.043	0.061	0.049
0.25	0.083	**0.029**	0.043	0.067	0.049
0.50	0.113	**0.033**	0.043	0.076	0.049
1.00	0.191	0.046	0.043	0.084	0.049
2.00	0.423	0.068	0.043	0.102	0.049

Naive is flat at 0.043 throughout, as it must be — it never touches O_0 — which makes it a clean control: everything that moves is attributable to the negative control's quality.

[N2] Validity: δ breaks Assumption 3.1

		Systematic bias		Total error	
δ	Recov. err	plug-in	naive	plug-in	naive
0.00	0.083	**0.029**	0.043	0.067	0.049
0.25	0.190	**0.032**	0.041	0.067	0.047
0.50	0.378	0.039	0.038	0.068	0.047
1.00	0.767	0.055	0.034	0.071	0.046

Caveats: δ implements one specific violation (leakage of the negative control's own noise into the reward); others may behave differently. The thresholds are properties of this environment, not universal constants — the transferable claim is the ordering, that validity matters more than strength. Naive's bias also drifts slightly downward as δ grows ($0.043 \to 0.034$), so the widening gap is not purely the plug-in degrading.

E. Corrections to Our Own Results

We record these because the verification process producing them is part of the contribution, not an embarrassment to be smoothed over.

Correction 1: harm ratio at zero confounding. On a 3-seed sweep we reported the model-based estimator as 23–46$\times$ worse than naive at $\kappa = 0$. At 5 seeds the ratio is 2.8–6.8$\times$. Still a real, clear negative result — just a substantially smaller one than we had stated.

Correction 2: retracted variance-reduction claim. We had claimed James–Stein shrinkage "cuts run-to-run variance sixfold under strong confounding." It does not replicate. Across the full 15-cell sweep, shrinkage has lower variance than the plain estimator in only $7/15$ cells — roughly a coin flip — and at $\kappa = 1$ specifically, the exact condition cited, its variance is *higher* at N=5,000 and N=20,000. The claim is retracted. The honest summary is a modest, noisy $\approx 7\%$ average MAE reduction, not a variance-reduction guarantee.

Neither was a coding bug: both were correctly computed standard deviations over an n=3 sample, which is too few draws to be stable. The lesson we take forward is that a claim derived from a *standard deviation* needs more seeds than a claim about a mean alone.

Correction 3: scope of the continuous study. We initially described the continuous extension as testing "the paper's own generality." Re-reading Hong et al. (2024) showed it assumes continuous S and O but an explicitly *finite* action space, while our environment uses a real-valued action. The oracle check still passes exactly, but the work is extrapolation beyond the stated theorem rather than an instantiation of it. We corrected the framing rather than relabel the experiment, and then ran the paper-faithful configuration (Section 4.6).

Correction 4: retracted mechanism for the selection failure. We had hypothesised that pessimistic selection underperformed because worst-case perturbations compound through the T=3 chain, citing coordinate-descent restart gaps that grew with region width. The finite-action experiment refutes this on two counts: the anomaly persists when the action space is finite, and restart gaps there are $\leq 4 \times 10^{-16}$ up to c=1, so the optimizer is finding the global minimum. The mechanism is instead behavior-policy coverage (Table 2). We report the failed hypothesis rather than silently substituting the new explanation, because the sequence — hypothesis, targeted experiment, refutation, replacement — is what makes the final claim credible.

F. Continuous Extension: Per-Policy De-biasing

Continuous kernel extension, 5 seeds, mean ± 95% CI at N=2,000 (summarised in Section 4.5). The pattern to note is in the intervals: naive bias is tight but systematically wrong, plug-in bias is several times smaller but noisier — the bias-for-variance trade that recurs throughout this paper.

Policy	Naive bias	Plug-in bias
never_treat	-1.255 ± 0.018	$\mathbf{-0.401 \pm 0.216}$
constant_dose	$+0.093 \pm 0.013$	$\mathbf{+0.008 \pm 0.032}$
proportional	-0.435 ± 0.013	$\mathbf{-0.129 \pm 0.061}$
aggressive	$+1.357 \pm 0.038$	$\mathbf{+0.365 \pm 0.178}$

G. Finite-Action Experiment: Full Results

Environment: continuous scalar state and observation, $A = \{0, 0.5, 1.0\}$, T=3, behavior policy a softmax over dose levels driven by the true hidden state (so overlap holds by construction). Candidate policies are observation-independent distributions over A; that restriction is what preserves the exact oracle, since the expected bridge feature under such a policy is exactly $[\pi, m_t]$ and the chain stays affine in the mean.

Two implementation improvements over Appendix B follow from the finite action space. The bridge feature map $[\mathrm{onehot}(a), o]$ is $(K{+}1)$-dimensional, so the primal solve is mandatory from the outset rather than discovered. And because the chain is affine in the mean, dynamics-block gradients become *exact* via the backward recursion $D_t = b_{R,t}[-1] + b_{D,t}[-1]\, D_{t+1}$, replacing the finite difference of Appendix B; verified against numerical differentiation to 1.16×10^{-10}.

N	300	800	2,000	4,000
Reward-bridge error	0.147	0.141	0.083	**0.071**
Dynamics-bridge error	0.271	0.251	0.156	**0.086**

Recovery is monotone in both families, cleaner than the continuous-action run, which was non-monotone at an intermediate point.

c	0.03	0.1	0.3	1.0	3.0
Fraction valid	1.00	1.00	1.00	1.00	1.00
Pessimistic regret	0.211	0.211	0.289	0.602	0.602
Plug-in regret	0.000	0.000	0.000	0.000	0.000
Max restart gap	0.0	1.1e-16	2.2e-16	4.4e-16	5.33

Plug-in selection is perfect at every width; pessimistic selection is not, and the near-zero restart gaps rule out an optimization explanation. See Table 2 for the coverage diagnostic that does explain it.

H. Bugs Found by Oracle Verification

Two implementation bugs in the model-free comparator were invisible from the derivation and surfaced only against exact ground truth. First, a per-stage direct-method implementation was silently biased off-policy: a policy with true value 1.865 evaluated to 1.61, with the error *not* shrinking in N — the fingerprint of a derivation error rather than noise. It was fixed with a proper backward recursion. Second, the corrected version naively scaled to the benchmark attempted to build and invert an $\approx 11{,}500 \times 11{,}500$ dense matrix per policy per fit ($\approx 2\,$GB), which stalled the run entirely; exploiting a block-diagonal structure the naive implementation missed resolved it.

The rank-deficiency issue of Appendix B belongs to the same category, with the additional wrinkle that it was invisible even from the downstream recovery metrics.

I. Correction: the Pessimism Numbers on a Common Grid

Three defects in our own implementation and two in our reporting were found after the results of Section 4.3 were first written up. All are corrected here.

The omitted class bound. Hong et al. (2024) minimise over $\mathrm{conf}(\alpha) \cap \mathcal{B}_{R,t}$, and the uniform bound M_R on that class carries through their Theorem 4.2. Our implementation minimised over the confidence region alone. This is an error on our side, not in the theory, and it inflates every pessimistic magnitude we reported.

The class carries two bounds, not one. Hong et al.'s Assumption 4.1(f) bounds the class in *sup*-norm; their Assumption D.16(d) additionally norm-constrains it in RKHS norm, which in this tabular instantiation is the coefficient ℓ_2 norm. Neither our ball nor our box alone is therefore the paper's literal class — each relaxes it, and the paper's class is their intersection, at least as strong as the stronger of the two at every width. We report both. The box alone is the weakest of the three: in this 36-dimensional coefficient space the tightest box containing the true bridge admits ℓ_2 norms up to 5.2 against the true bridge's 1.803.

Cross-c reporting. Section 4.3 quoted vanilla at c=0.1 against the projection at c=0.03. On a common grid the contrast disappears.

Toy POMDP, T=3, N=20,000, 3 seeds, $V_{\mathrm{true}} = 2.0177$. Selection regret on the left, V_{low} on the right:

	Selection regret				V_{low}			
c	0.03	0.10	0.30	1.00	0.03	0.10	0.30	1.00
Plug-in	**0.000**	**0.000**	**0.000**	**0.000**	2.052	2.052	2.052	2.052
Vanilla	0.000	0.257	0.513	0.513	1.272	0.123	−3.235	−20.983
ℓ_2 ball	0.000	0.257	0.513	0.513	1.278	0.469	−0.806	−3.026
Box (4.1(f) only)	0.000	0.257	0.427	0.427	1.275	0.155	−2.729	−10.812
Signal-projected	0.000	**0.000**	0.257	0.513	1.338	0.813	−0.298	−4.015

What this changes. At c=0.03 — the operating point at which Section 4.3 reported the projection's 0.000 — *every* method including vanilla achieves 0.000, so the published advantage is an artifact of comparing across region sizes. The projection does have a real advantage in the middle of the grid: at c=0.1 it alone among the pessimistic variants still selects the optimal policy, and at c=0.3 its regret is half the others'.

What this does not change. The ranking failure of vanilla pessimism at larger c survives the class correction, and the null-space mechanism of Section 3.3 is untouched. Identification, estimation and de-biasing results do not involve the pessimism layer.

The baseline dominates. The confounding-aware plug-in achieves 0.000 regret at every width tested and is never beaten by any pessimistic variant. On this environment the pessimism layer costs selection accuracy and returns nothing. We report this alongside the projection's 0.000, because a pessimistic method that does not beat its own plug-in has not earned its complexity here.